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1. Introduction and principle

The soil-moisture sensor node is based on the principle of capacitive sensing: the dielectric permittivity of soil changes with its water content. This scientific principle is well established. The proposed application is automatic irrigation scheduling for small rice farms.

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2. Laboratory results

A breadboard of the sensing circuit was tested in the laboratory with five soil samples. Laboratory measurements showed a monotonic response between 5 and 45 percent volumetric water content. The proof-of-concept met its goal of detecting a 5 percent change.

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3. Field-like testing

A representative prototype was demonstrated in a relevant environment: an environmental chamber at 35 C and 80 percent humidity. The test conditions and the configuration of the prototype are listed in Table 2. Measured accuracy was within the specified tolerance of 3 percent. The acceptance criteria were met.

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4. Summary

The management summary states that the system has reached TRL 6. No pilot installation in a farm has been carried out yet.